

Jacob Vaughn
Name

ELCT 432 - Quiz 2

Instructions: Please show ALL work and use clear English to explain methods of solution. Unclear or illegible work will not receive full credit.

Honor Pledge: All work on this quiz is my own, and I have not received any unauthorized aid.

Signature:

Jacob Vaughn

1) A baseband signal $x(t)$ has spectrum that is non-zero only for $|f| < 10$ MHz. For some arbitrary shape $X(f)$,

a) (10 pts) Sketch $X(f)$

b) (40 pts) Sketch the spectrum of $y(t) = e^{i2\pi f_c t} x(t)$, for $f_c = 2.4$ GHz

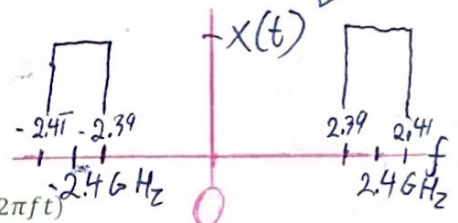
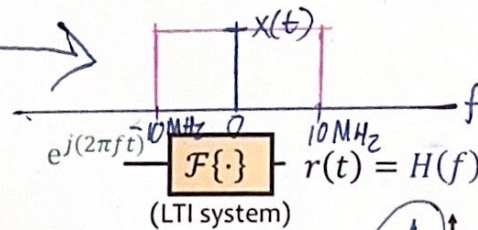
2) In free space, the propagation delay for electromagnetic waves can be calculated as $\tau = d/c$ in second, where c is the speed of light (approximately equal to 3×10^8 meters/sec in the free space) and d is the distance between transmitter and receiver in meters.

Calculate the frequency response for the line-of-sight path at 1 GHz for the scenarios below:

a) (25 pts) $d = 30$ meters

b) (25 pts) $d = 30.15$ meters = 30 meters + 0.15 meters.

Hint:



$$\textcircled{a} \quad \tau = \frac{30}{3 \times 10^8} = 1 \times 10^{-7}$$

$$= e^{j2\pi \cdot 10^9 \cdot 10^{-7}} \Rightarrow e^{-j2\pi}$$

a)

$$\textcircled{b} \quad \tau = \frac{30.15}{3 \times 10^8} = 1.005 \times 10^{-7}$$

$$= e^{-j2\pi \cdot 10^9 \cdot 10^{-7} \cdot 1.005}$$

$$= e^{-j2.01\pi}$$

b)